

Hassan Ali Shah

Full Stack Developer | Software Engineer | Agentic AI Engineer

Lahore, Pakistan | +92 325 4331778 | bscs24040@itu.edu.pk | github.com/Hassan-Ali-17 | linkedin.com/in/hassan-ali-shah

Portfolio : [Hassan Ali Portfolio](#)

PROFESSIONAL SUMMARY

Ambitious Full Stack Developer and Agentic AI Engineer with a strong foundation in Data Structures, Algorithms, and scalable system design. Experienced in building production-ready web applications and implementing machine learning models from scratch. Passionate about Generative AI and autonomous systems, with hands-on industry experience at an AI startup. **Internationally recognized innovator** — represented Pakistan at the global stage in **Act in Space 2026**, organized by **CNES (French Space Agency)** and the **European Space Agency (ESA)**. Proven leader with the ability to drive teams, solve complex problems, and rapidly adapt to emerging technologies.

TECHNICAL SKILLS

Languages	C++, Python, JavaScript, SQL
Web Development	React.js, Node.js, MongoDB, REST APIs (MERN Stack)
AI / ML	Generative AI, Agentic AI Systems, Logistic Regression, NumPy, Data Preprocessing
Core CS	Data Structures & Algorithms, OOP, System Design, Dynamic Memory Management

PROFESSIONAL EXPERIENCE

ML/AI Engineering Intern | CAASIAC | 2024 – 2025

Contributed to end-to-end machine learning workflows at an AI-driven startup, gaining hands-on experience in a fast-paced production environment.

- Designed and optimized data preprocessing pipelines to improve model input quality and training efficiency.
- Developed and evaluated ML models to support AI-driven problem-solving across business use cases.
- Collaborated with senior engineers on agentic AI research and implementation strategies.
- Gained practical exposure to Generative AI tools and workflows in a real-world product environment.

PROJECTS

Full Stack Web Application (MERN Stack)

Architected and deployed a production-ready web application featuring JWT-based authentication, dynamic real-time data flow, and a fully responsive UI using React.js, Node.js, Express, and MongoDB.

Machine Learning from Scratch – Logistic Regression

Implemented a complete logistic regression classifier from the ground up using NumPy, including gradient descent optimization, loss tracking, and performance evaluation — without high-level ML libraries.

Cricket Team Management System (C++)

Built a feature-rich management system in C++ using raw arrays, pointers, dynamic memory allocation, and file I/O — deliberately avoiding STL to demonstrate deep understanding of low-level programming.

Algorithm Performance Benchmarking

Designed and analyzed recursive vs. iterative algorithmic solutions, profiling time and space complexity to identify and optimize performance bottlenecks in real-world scenarios.

ACHIEVEMENTS & RECOGNITION

■ National Winner — Act in Space 2026

Secured 1st place at the national level in Act in Space 2026, a prestigious space innovation challenge.

■ International Representative — Act in Space 2026 (France)

Selected to represent Pakistan at the international finals of Act in Space 2026, organized by **CNES (Centre National d'Études Spatiales)** and the **European Space Agency (ESA)** — one of the few Pakistanis to compete at this global platform.

■ **Team Lead — NASA Space Apps Challenge**

Led a team in the world's largest global hackathon organized by NASA, directing technical strategy and project delivery under a tight international competition timeline.

■ **Participant — Code Rush Hackathon**

Competed in a high-intensity coding competition, applying problem-solving and algorithmic skills under pressure.

EDUCATION

BS Computer Science Information Technology University (ITU), Lahore	2024 – 2028 CGPA: 3.02 / 4.0
FSc Pre-Engineering Punjab Group of Colleges, Lahore	2022 – 2024 87%
Matric (Science) Lahore Grammar School	2020 – 2022 92%

CORE STRENGTHS

- Strong analytical and problem-solving skills with a track record of competitive achievement
- Ability to build robust systems from first principles — hardware-level to cloud-deployed applications
- Proven leadership in high-pressure, international-level hackathon environments
- Fast learner who thrives at the intersection of cutting-edge AI research and practical engineering
- Deep grasp of core Computer Science fundamentals: algorithms, memory management, and system architecture



Portfolio

Github

Linkedin